

Aanal Jayesh Shah

 Google Scholar  ORCID  LinkedIn  Email

EDUCATION

PhD in Physics and Astronomy

January 2021 - August 2026 (Expected)

Purdue University

B.E. Electronics and Communications Engineer

2015 - 2019

L.D. College of Engineering

PUBLICATIONS

Aanal J. Shah, Peter Kirton, Simone Felicetti, Hadiseh Alaeian
Dissipative Phase Transition in the Two-Photon Dicke Model

Phys. Rev. Lett. **135**, 173602 (2025)

Lijuan Dong, **Aanal J. Shah**, Peter Kirton, Hadiseh Alaeian, Simone Felicetti
Collective Enhancement of Photon Blockade via Two-Photon Interactions

[arXiv:2511.11506](#) (2025)

[Accepted for publication in PRX Quantum](#)

Preprints & In Preparation

Aanal J. Shah*, Alessandro Coppo*, Roberto di Candia, Hadiseh Alaeian, Simone Felicetti (in preparation)
Nonlocal Quantum Phase Transitions

[To be submitted to Nature Physics](#)

Aanal J. Shah, Lijuan Dong, Peter Kirton, Hadiseh Alaeian, Simone Felicetti
Squeezing in the Standard and Two-Photon Tavis-Cummings Model

(in preparation)

RESEARCH EXPERIENCE

Squeezing, Non-Classical & Non-Gaussian Light Generation Schemes | *Purdue*

2025 – present

- Showed that the two-photon Tavis-Cummings model produces robust squeezing in the thermodynamic limit (many atoms), and enabling scalable non-classical light generation.
- Computed photon emission rates and multi-time field correlations $g^{(1)}(t_1, t_2)$ and $g^{(2)}(t_1, t_2)$; characterized emitted field via Wigner functions and photon statistics.
- Results establish a route to **on-demand squeezed light sources** for continuous-variable (CV) QKD and quantum-enhanced sensing applications.

Nonlocal Quantum Phase Transitions via Entangled Environment | *Sapienza University*

2025 – present

- Collaborated with Prof. Simone Felicetti on understanding critical phase transitions are driven by nonlocal quantum fluctuations.
- Implemented Fokker-Planck equations for analytical calculation. Worked with Lipkin–Meshkov–Glick (LMG) model and two-photon driven Kerr resonators.
- Contributions: theoretical modelling, derivation of the Lindblad Master Equation, numerical calculations using quantum Monte Carlo, manuscript writing.

Collective Interactions using Non-linear Quantum Optics | *PhD Thesis, Purdue University*

2023 – present

- Published work [Phys.Rev.Lett.](#) characterizing the driven-dissipative phase transition of the two-photon Dicke model, including finite-size scaling and beyond-mean-field quantum fluctuation effects.
- Demonstrated **collective photon blockade enhancement** in two-photon interacting atomic ensembles [arXiv:2511.11506](#), with implications for deterministic single-photon sources and quantum gate fidelity.
- Independently built all numerical simulation codes in Python, Julia, and Mathematica; executed large-scale parallel jobs on HPC infrastructure.

Pattern Formation in Exciton-Polariton Condensates | *Purdue University*

2024 – present

- Modeled coherent drive modulation in Rydberg exciton-polariton condensates (Cu_2O) using the Gross-Pitaevskii equation and Hydrodynamic formulation; applied Floquet analysis and Lyapunov exponents to map stability and understand the jump behavior between stable branches.

Rydberg Tweezer Array Lab | *Purdue University*

2021 – 2022

- Contributed to starting up a Rydberg atom AMO lab from scratch: aligned Rb laser optics, **designed 4-layer PCBs** (Eagle) for pulse and magnetic coil control, and built NVIDIA-based hardware/software control architecture.

Quantum Key Distribution – BB84 Protocol | *Physical Research Laboratory, India*

2019 – 2020

- Designed and built hardware for a BB84 fiber-optic QKD testbed: laser driver circuit, single-photon detection electronics, and optical channel interface.
- Implemented **LDPC error-correction** in MATLAB via Monte Carlo simulation; optimized code parameters to maximize distilled key length at low QBER.
- Proposed a novel receiver architecture reducing intercept-resend eavesdropping efficiency; developed an **alternative to decoy states** that simultaneously improved QBER and raw key rate.
- Contributed both theoretical security analysis and circuit-level hardware implementation.

Aditya Tokamak Automation | *Institute for Plasma Research, India - Bachelor's Thesis*

2018 – 2019

- Automated high-voltage capacitor charge/discharge for Tokamak magnetic coils via LabVIEW and PLC communication over TCP/IP; implemented secure data logging with access control.

AWARDS

Charlotte Ida Litman Tubis Award

2026

Summer Research Grant

2023

CONFERENCES AND PRESENTATIONS

Les Houches Postdoctoral School on Photons and Atoms

2026

SPARK Exotic Quantum Matter – San Sebastian

2026

APS Division of Atomic, Molecular and Optical Physics (DAMOP)

2025 & 2026

Purdue Physics AMO Seminar

2025

Invited talk- Physical Research Laboratory (PRL), Ahmedabad

2024

Midwest Cold Atom Workshop (MCAW)

2024

- Presented work on Superradiance in the two-photon Dicke model.

Midwest Quantum Collab Workshop (MCQ)

2023

- Presented work on Generation of Non-classical States Using Rydberg Interactions.

TEACHING EXPERIENCE

PHYS 214: The Nature of Physics – Instructor

Summer 2021

- Designed course structure, taught lectures, and graded assignments for this course.

PHYS 233/234: Physics for Life Sciences – Teaching Assistant

2022-2024

- Supported 200+ undergraduate students per semester through teaching recitations, labs, grading, and office hours.

SKILLS

AMO & Quantum Optics: Cavity QED, cold atoms / BECs, Rydberg arrays, Open quantum systems, Lindblad master equations, cavity QED, driven-dissipative phase transitions, photon blockade, superradiance / subradiance, squeezing & non-classical light, many-body physics, mean-field & Bogoliubov methods, Floquet analysis, quantum entanglement, nonlocality

Quantum Information & Communication: Quantum key distribution (BB84), LDPC error-correction codes, quantum bit error rate (QBER) analysis, decoy-state protocols, fiber-optic quantum channels, continuous-variable (CV) QKD, entanglement-based communication

Programming & HPC: Python (QuTiP, NumPy, SciPy, Matplotlib), Julia, MATLAB, C / C++ , Mathematica, LabVIEW, Eagle, MPI / parallel computing, SLURM scripts

Languages: English (Proficient), Gujarati (Proficient), Hindi (Proficient), Italian (CEFR A2 level)

REFERENCES

- **Dr. Hadiseh Alaeian**
Assistant Professor
Electrical and Computer Engineering and Physics and Astronomy
Purdue University
halaeian@purdue.edu
- **Dr. Peter Kirton**
Professor
Department of Physics
University of Strathclyde
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- **Dr. Simone Felicetti**
Research Scientist
Institute for Complex Systems
CNR – Italy and Sapienza University
simone.felicetti@cnr.it